

PHILOSOPHY 130: PHILOSOPHY OF SPACE AND TIME
ASSIGNMENT 2: GEOMETRY AND CONVENTIONALISM

Due: April 14, in class

Name: _____

Please draw and write your answers in the space provided.

1. Geometry on a sphere. In the space below, sketch a sphere and draw the following on its surface:

- (a) Two “straight lines” (great circles) that start out appearing parallel at the equator but eventually intersect. Label the intersection point.
- (b) Two triangles on the sphere: a *small* triangle (covering a tiny patch of the surface) and a *large* triangle (e.g., with one vertex at the pole and two on the equator).
- (c) Next to each triangle, write an estimate of whether its angle sum is close to 180° or significantly greater. In a sentence or two, explain why the size of the triangle matters for detecting non-Euclidean geometry.

2. Poincaré's heated disk. Huggett (Ch. 13) and Norton both describe a thought experiment involving inhabitants of a disk-shaped world where measuring rods shrink as they approach the edge.

- (a) In the space below, draw a large circle representing the disk. Inside it, draw several copies of a measuring rod at different distances from the center—showing how the rod gets shorter near the boundary. Then draw the path that an inhabitant would consider a “straight line” (it will appear curved in the representation in the disk, bending away from the edge).
- (b) In 2–3 sentences, explain why the “straight lines” followed by the inhabitants *appear curved* in the representation in the disk: what is it about the shrinking of the rods near the boundary that makes the shortest measured path bend away from the edge?

3. Intrinsic vs. extrinsic geometry. Imagine you are a two-dimensional being living on some surface. You have a measuring rod and a protractor but cannot “see” the surface from outside—you have no access to any embedding space. Describe in 2–3 sentences a procedure you could carry out, using only measurements available to you on the surface, to determine whether your surface is flat, positively curved (like a sphere), or negatively curved (like a saddle). Be specific about what you would measure and what result would indicate each case. (You may also find it useful to draw a diagram illustrating your procedure.)